# Giuseppe Floris

PhD Student in Electronics and Computer Engineering

University of Cagliari, Italy

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## Research profile

I am a member of sAIferLab and a PhD student in Electronics and Computer Engineering at the University of Cagliari. My research focuses on machine learning security and privacy, and on applications of machine learning to cybersecurity. In particular, I work on adversarial robustness and privacy attacks and defenses. I am interested in evaluating security and privacy risks in machine learning systems and developing methods to mitigate them.

## Research experience

**Visiting PhD** (Supervisor: Prof. Franziska Boenisch) Nov 2025 - Apr 2026

CISPA Helmholtz Center for Information Security, Germany · SprintML Lab

Research on privacy attacks and defenses in LLMs and their tokenizers.

**Research Intern** Jan 2025 - Jul 2025

PLURIBUS ONE S.r.l.

Development of API-Sec, an ML-based threat detection framework for web APIs.

**Member · sAIfer Lab** 2024 - present

University of Cagliari, Italy

Member as a PhD Student of the sAIfer Lab at the Department of Electrical and Electronic Engineering.

**Research Grant** Jun 2023 - Sep 2023

University of Cagliari, Italy

Comparative analysis of robustness of image classifiers against state-of-the-art perturbation models using PyTorch.

## Education

**PhD in Electronics and Computer Engineering** 2023 - 2026

University of Cagliari, Italy · DRIEI PhD Program

Supervisor: Prof. Battista Biggio · Co-supervisor: Dr. Maura Pintor.

**MSc in Computer Engineering, Cybersecurity and Artificial Intelligence** 2021 - 2023

University of Cagliari, Italy

*110/110 with honors.* Thesis: *Hyperparameter Optimization for Fast Minimum-norm Attacks*.
Supervisor: Prof. Battista Biggio.

**BSc in Electrical, Electronic and Computer Engineering** 2016 - 2020

University of Cagliari, Italy

Thesis: *Mitigation of vulnerabilities in the evolution of open source software development*.
Supervisor: Prof. Giorgio Giacinto.

## Publications under review

- **Privacy attacks and DP protections in Large Language Models' tokenizers.**
- **Benchmarking privacy risk on LLMs.**

## Publications

### ModSec-AdvLearn: Countering Adversarial SQL Injections With Robust Machine Learning

**Giuseppe Floris**, Christian Scano, Biagio Montaruli, Luca Demetrio, Andrea Valenza, Luca Compagna, Davide Ariu, Luca Piras, Davide Balzarotti, Battista Biggio.

IEEE Transactions on Information Forensics and Security (TIFS), vol. 20, pp. 6693-6705, 2025.

### HO-FMN: Hyperparameter Optimization for Fast Minimum-Norm Attacks

Raffaele Mura, **Giuseppe Floris**, Luca Scionis, Giorgio Piras, Maura Pintor, Ambra Demontis, Giorgio Giacinto, Battista Biggio.

Neurocomputing, vol. 616, article 128918, 2025.

### ModSec-Learn: Boosting ModSecurity with Machine Learning

Christian Scano, **Giuseppe Floris**, Biagio Montaruli, Luca Demetrio, Andrea Valenza, Luca Compagna, Davide Ariu, Luca Piras, Davide Balzarotti, Battista Biggio.

21st International Conference on Distributed Computing and Artificial Intelligence (DCAI), Salamanca, Spain, 26-28 June 2024.

### Improving Fast Minimum-Norm Attacks with Hyperparameter Optimization

**Giuseppe Floris**, Raffaele Mura, Luca Scionis, Giorgio Piras, Maura Pintor, Ambra Demontis, Battista Biggio.

31st European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN), Bruges, Belgium, 4-6 October 2023.

## Technical skills

**Programming:** Python (AI/ML and LLM libraries).

**Machine learning & AI security:** applications to cybersecurity, hyperparameter optimization, adversarial attacks and robustness, privacy attacks, differential privacy.

## Languages

**Italian:** native **English:** B2

## Additional information

**Citizenship:** Italian.